FINETUNED LANGUAGE MODELS ARE ZERO-SHOT LEARNERS

Olcay Değirmenci
Student ID: 323690006

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Paper methodology

Objective. The FLAN paper investigates whether large language models can improve their zero-shot generalization by finetuning on a large collection of natural language instructions instead of task-specific labeled data.

Method. The authors create a large instruction-tuning dataset by dividing 62 different NLP datasets into 12 task clusters that combine similar tasks (NLI, QA, translation, summarization, sentiment, etc.). Approximately 10 natural language instruction templates were written for each dataset; most of these describe the original task, while others are creative instructions that "reverse" the task (e.g., generating movie reviews instead of classification in sentiment analysis). In the finetuning phase, the model treats randomly selected examples from this mix as input-output pairs given entirely by natural language instructions.

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Training Setup & Model

Base Model. FLAN starts from LaMDA-PT, a 137B-parameter pretrained language model.

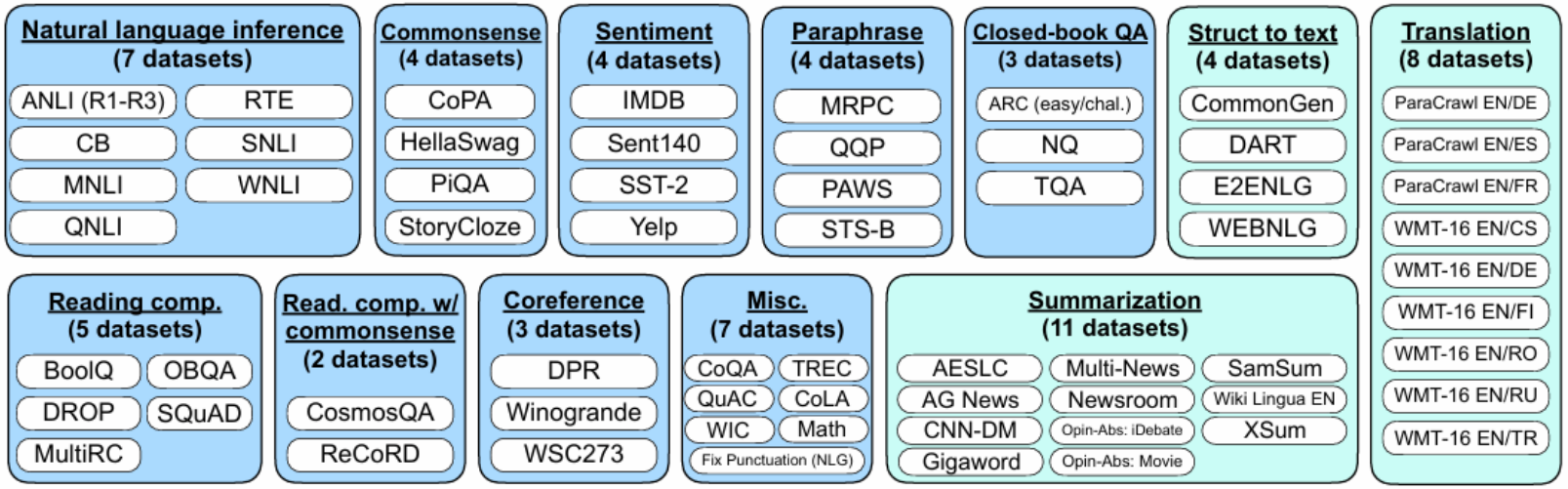
Instruction Tuning Procedure.

- 62 NLP datasets into task clusters (e.g., NLI, reading comprehension, QA, translation).
- Manually write diverse natural-language instructions and target formats for each dataset.
- Sample mixtures of tasks and instructions during finetuning to encourage broad generalization.

Model Validation Approach.

- Zero-shot and few-shot performance on held-out tasks not seen during instruction tuning.
- Comparisons against GPT-3, LaMDA-PT and other major models without instruction tuning.

Instruction-Tuning Datasets



Researchers prepared a large instruction-tuning dataset for 12 different task sets, including natural language extraction, reading comprehension, logical reasoning, sentiment analysis, paraphrasing, question-answer, summarization, and translation, by combining a total of 62 datasets.

Source: Wei et al. (2022).

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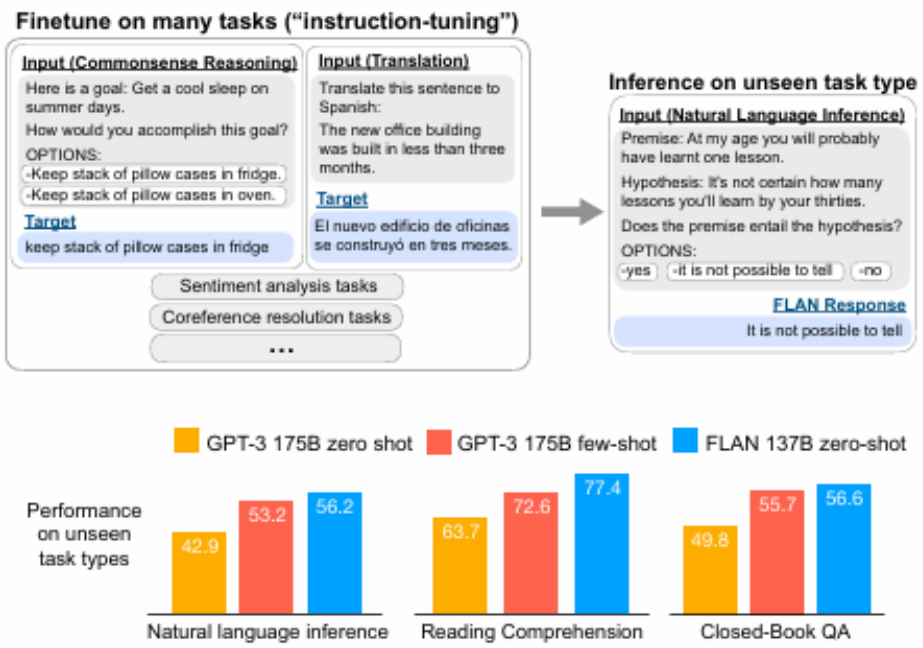
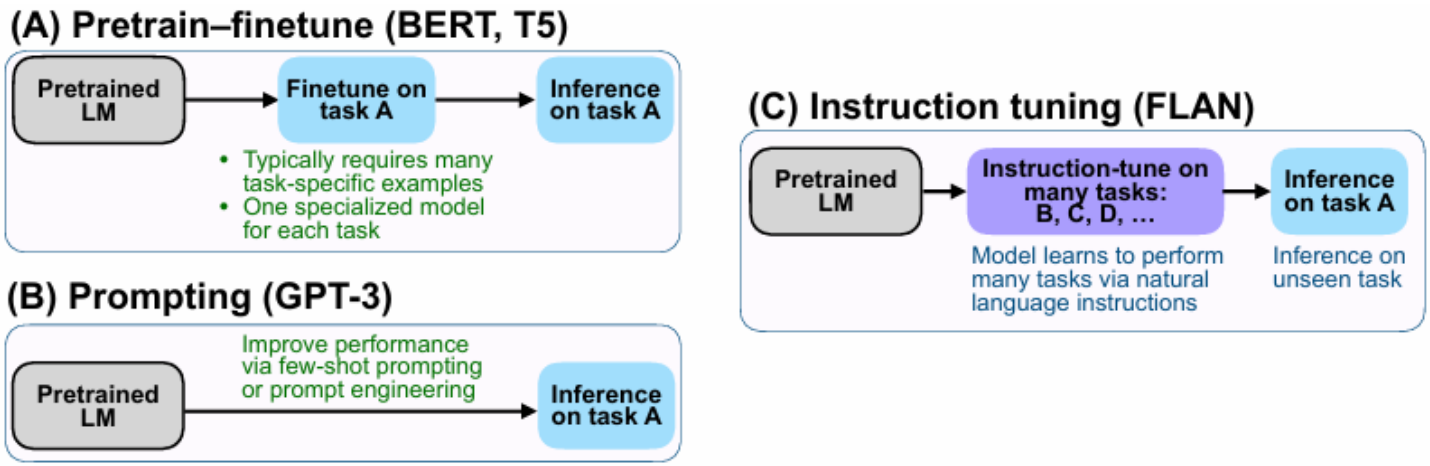
Zero-Shot Performance Across Task Clusters



Study principle: The model is first trained by fine-tuning natural language instructions on many different tasks, thus learning the instruction formats for a wide variety of tasks, such as translation, reasoning, and sentiment analysis. In the inference phase, the model performs a zero-shot correct response to a new task type it has never seen in training, simply by giving it instructions.

Source: Wei et al. (2022).

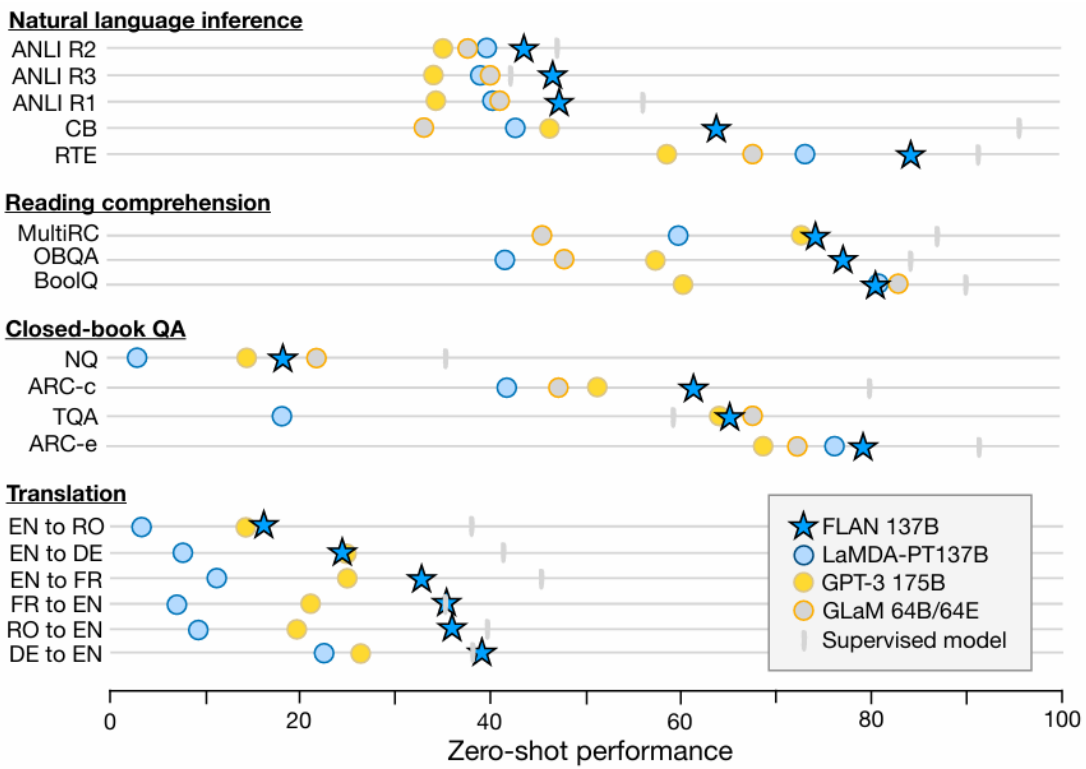
Three Paradigms of Language Model Use



Comparison of three approaches: (A) In the traditional pretrain–finetune method, the model requires fine-tuning with separate examples for each task; (B) The prompting approach (GPT-3) uses only a few examples or well-written prompts for tasks; (C) Instruction tuning (FLAN) trains the model with natural language instructions for tasks, allowing the model to perform better on both these tasks and new tasks it has never seen before, as a zero-shot.

Source: Wei et al. (2022).

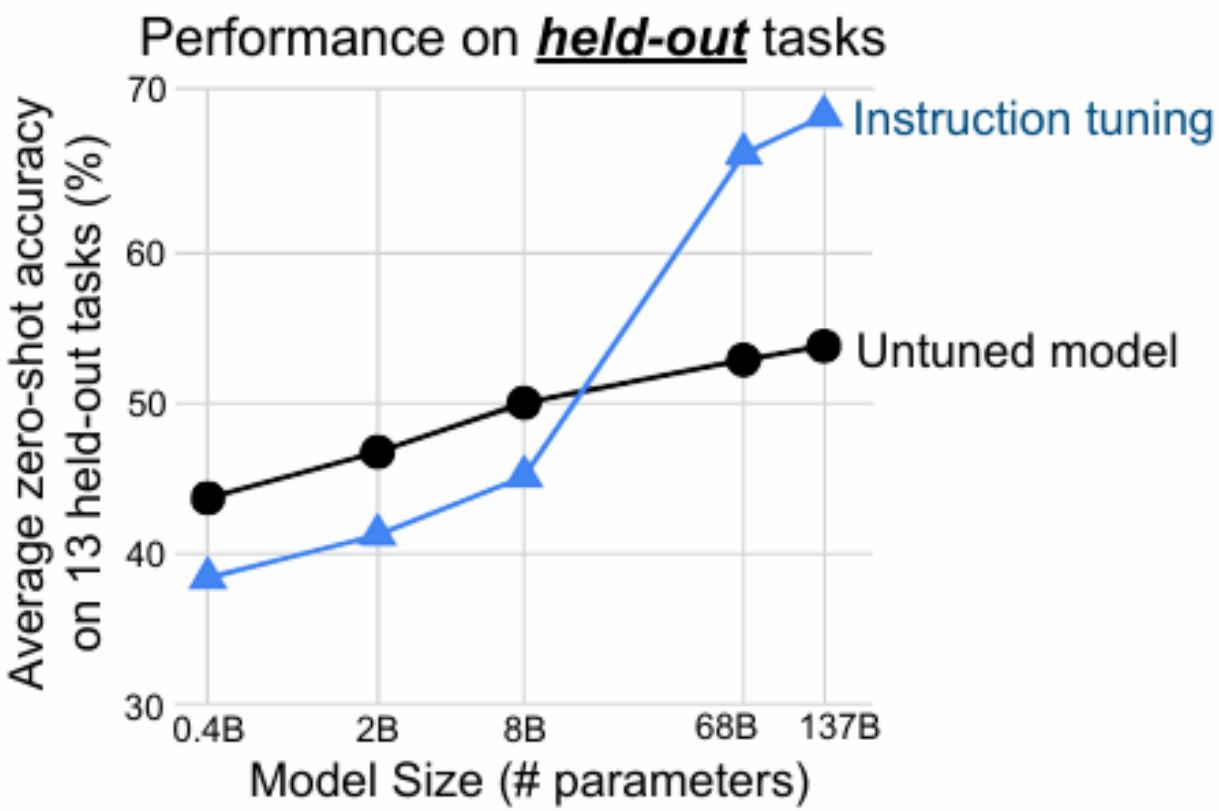
FLAN vs other models on Various Tasks



This graph shows that FLAN 137B outperforms other major models (GPT-3, LaMDA-PT, GLaM) on various tasks. FLAN achieves the best results on different comparison.

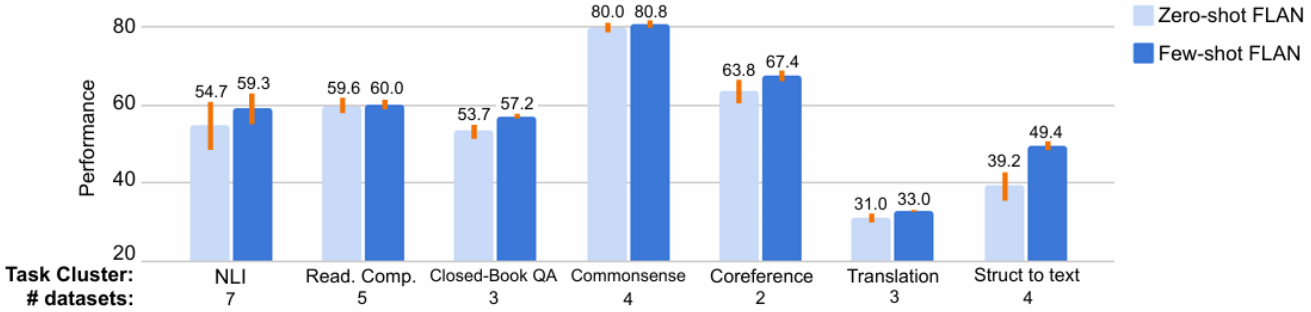
Source: Wei et al. (2022).

Scaling Effect on Held-Out Zero-Shot Tasks



As the model size increases, the difference in performance between untuned and instruction-tuned models increases. Instruction tuning becomes even more effective at larger scales. *Source: Wei et al. (2022).*

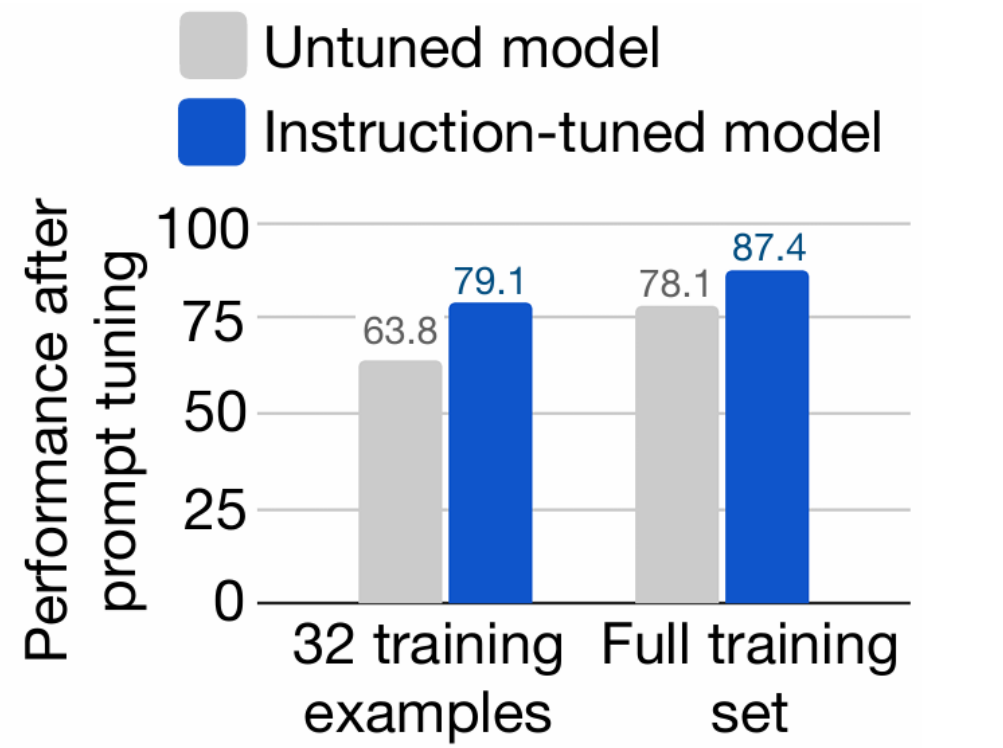
Zero-Shot vs Few-Shot FLAN Performance



Across several task clusters, few-shot prompting on top of an instruction-tuned model yields additional gains, confirming that instruction tuning provides a strong base for both zero-shot and few-shot learning..

Source: Wei et al. (2022).

Prompt Tuning: Tuned vs Untuned Models



Instruction-tuned models demonstrate improved adaptability, responding significantly better to fast tuning, both in extremely low-data scenarios (32 examples) and on full training sets.

Source: Wei et al. (2022).

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Key Findings

Key Findings. Zero-shot performance is significantly improved over the baseline model, LaMDA-PT 137B. FLAN outperforms GPT-3 175B on 20 out of 25 datasets. It even outperforms GPT-3 on many tasks such as ANLI, RTE, BoolQ, ARC, and OpenBookQA. Performance increases with task diversity; the number of instruction sets directly increases FLAN's generalization power to unseen tasks. The effect of instruction tuning is only positive for large models.

Impact. The study shows that the success of large models comes not only from increasing model size, but also from revealing a fundamental capacity for alignment with natural language instructions. In summary, scaling alone is not enough; models must be trained with the right instructions.

Limitations

Limitations

- The pretrained model (LaMDA-PT) and the large-scale dataset used for FLAN are not fully open-source, limiting independent reproducibility and verification.
- Strong performance relies on very large model sizes and high computational cost, restricting usage to resource-rich institutions.
- Instruction quality depends on manually written templates; bias and limited linguistic diversity may affect real-world generalization.

Ethical & Societal Aspects

Ethical & Societal Aspects

- Instruction tuning may amplify existing dataset biases (toxicity, stereotypes), as models become more responsive to biased or harmful instructions.
- Over-reliance on black-box large models in high-stakes settings raises concerns due to lack of interpretability and uncertain safety guarantees.
- Large models incur significant environmental and energy costs during pretraining and finetuning.

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Related Work

References

- Wei, J., et al. (2022). Finetuned language models are zero-shot learners. *arXiv preprint arXiv:2109.01652*.
- Aghajanyan, A., Gupta, A., & Zettlemoyer, L. (2021). MUPPET: Massive multi-task representations with pre-finetuning. *arXiv preprint arXiv:2101.11038*.
- Brown, T. B., et al. (2020). Language models are few-shot learners. *arXiv preprint arXiv:2005.14165*.
- Chung, H. W., et al. (2022). Scaling instruction-finetuned language models. *arXiv preprint arXiv:2210.11416*.
- Ouyang, L., et al. (2022). Training language models to follow instructions with human feedback. *arXiv preprint arXiv:2203.02155*.
- Sanh, V., et al. (2021). Multitask prompted training enables zero-shot generalization. *arXiv preprint arXiv:2110.08207*.

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Acknowledgment

Models such as ChatGPT and Gemini were used for language refinement and slide organization using Latex. The preparation and evaluation of the slide outlines were my own work. All figures used come from the published article by Wei et al. (2022).

Thank you for listening!